INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

# COAT, ALL-WEATHER, WOMAN'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a woman's all-weather coat.

1.2 Classification. The coat is available in the following types, classes, and sizes, as specified (see 6.2).

1.2.1 Types.

Type I – With right and left shoulder rifle patches

Type II – With right shoulder rifle patch only

1.2.2 Classes.

Class 1 – Pewter 2246

Class 2 – Black 385

Class 3 – Heritage Green 564

1.2.3 Sizes. The coat will be of the sizes specified in table I.

TABLE I. Schedule of sizes.

| Short | Regular | Long | Extra Long |
|---|---|---|---|
| X0 | X0 | X0 | -- |
| 2 | 2 | 2 | -- |
| 4 | 4 | 4 | -- |
| 6 | 6 | 6 | -- |
| 8 | 8 | 8 | -- |
| 10 | 10 | 10 | -- |
| 12 | 12 | 12 | -- |
| 14 | 14 | 14 | -- |
| 16 | 16 | 16 | 16 |
| 18 | 18 | 18 | 18 |
| 20 | 20 | 20 | 20 |
| -- | 22 | 22 | 22 |

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8410

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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50198 - Thread, Gimp, Cotton, Buttonhole

A-A-50199 - Thread, Polyester Core, Cotton or Polyester-Covered

A-A-50531 - Cloth, Poplin: Polyester and Cotton (Water Repellent)

A-A-55212 - Belt, All-Weather Coat, Women's

A-A-55634 - Zippers (Fasteners, Slide Interlocking)

A-A-59994 - Button, Sewing Hole, and Button, Staple, (Plastic)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1391 - Provisions for Evaluating Quality of Overcoats, Men's

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Gray Scale for Color Change

AATCC EP9 - Visual Assessment of Color Difference of Textiles

AATCC TM8 - Colorfastness to Crocking: Crockmeter Method

AATCC TM15 - Colorfastness to Perspiration

AATCC TM20 - Test Method for Fiber Analysis: Qualitative

AATCC TM22 - Water Repellency: Spray Test

AATCC TM27 - Test Method for Wetting Agents: Evaluation of Rewetting Agents

AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated

AATCC TM70 - Water Repellency: Tumble Jar Dynamic Absorption

AATCC TM81 - pH of the Water-Extract from Wet Processed Textiles

AATCC TM96 - Dimensional Changes in Commercial Laundering

AATCC TM118 - Oil Repellency: Hydrocarbon Resistance Test

AATCC TM127 - Test Method for Water Resistance: Hydrostatic Pressure

AATCC TM132 - Colorfastness to Drycleaning

AATCC TM158 - Dimensional Changes on Drycleaning, Perchloroethylene

AATCC TM164 - Colorfastness to Oxides of Nitrogen Under High Humidities

(Copies of these documents are available online at www.aatcc.org.)

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM D737 | - | Standard Test Method for Air Permeability of Textile Fabrics |
| ASTM D1424 | - | Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus |
| ASTM D3775 | - | Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics |
| ASTM D3776/D3776M | - | Standard Test Methods for Mass Per Unit Area (Weight) of Fabric |
| ASTM D5034 | - | Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test) |

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Description.

3.3.1 Coat. The coat (see figures 1 through 4) is made with a water-repellent-treated-material (see 3.4.1) shell. The coat design is a double-breasted front with a 6-button closure closing right over left, a convertible collar with button closure, a belt with buckle, a two-piece back with a center vent with a single button right over left closure, a back yoke, split raglan sleeves, button-down pointed shoulder straps, button-down pointed sleeve straps, and a half lining at the front and at the back with fully lined sleeves. The coat front has two vertical welt pockets with pass-through slits, a 1/4-inch-wide hanger loop at back neck with a minimum finished opening of 1-7/8 inches, and extra buttons sewn on left front facing. The coat has a removable liner (see figures 5 and 6) with pass-through slits coordinated with the pocket slits in the coat. The type I coat has right and left front shoulder flaps (rifle patches) and is worn primarily by the United States Marine Corps; the type II coat has only a right front shoulder flap (rifle patch) and is worn primarily by the United States Army and the United States Navy.

3.3.1.1 Belt and buckle. The belt and buckle (see 3.5.4) is 2-1/8±1/8 inches wide and has a buttoned belt anchor with a belt loop keeper, fused interlining, and a buckle.

3.3.1.2 Belt loops. The coat has four belt loops, finishing 2-5/8±1/8 inch in length and 3/8 to 1/2 inch wide. The loops are securely bartacked at the top and bottom of the loop to form an opening of at least 2-1/4 inches. The belt loop on the right-side seam has an under loop for the belt anchor. The loops are positioned as follows:

a. One loop on each front, positioned vertically in line with front corner of the front pocket flap seam line and the base of the loop is positioned 1/4±1/16 inch above the highest point of the pocket flap.

b. One loop positioned vertically on each side seam and in horizontal alignment with the front loops.

3.3.1.3 Shoulder straps. The shoulder straps are centered along the shoulder seam and the pointed end finishes even to 1/2 inch from the collar setting seam. The shoulder straps finish smooth and flat without distortion.

3.3.1.4 Buttons. The coat has buttons in the following locations:

a. Four 45 ligne buttons on the left front to mate with buttonholes on the right front.

b. Four 45 ligne buttons on the right front in alignment with the buttonholes on the right front.

c. One 30 ligne button on each shoulder to mate with buttonholes on the epaulets.

d. One 30 ligne button on each sleeve cuff positioned to mate with buttonholes on the sleeve tabs.

e. One 24 ligne button inside each sleeve cuff positioned to mate with button loops on the removable liner sleeve.

f. One 24 ligne button inside the left vent to mate with the buttonhole inside the right vent.

g. Four 18 ligne buttons inside the right front, one backing each of the 45 ligne buttons.

h. One 45 ligne button and one 30 ligne button sewn on the lower inside left facing, as spares.

3.3.1.5 Sleeve straps. The sleeve straps are positioned on the back seam of the sleeve according to notches on the pattern. The pointed end of the sleeve strap finishes even to 1/2 inch from the sleeve top seam. The sleeve strap is parallel to the bottom edge of the sleeve. The sleeve straps finish smooth and flat without distortion.

3.3.2 Removable liner. The removable liner has bound edges, set-in sleeves, two bound pass-through pocket slits and a center back vent. The liner's sleeve bottom is bound, with a button loop on the underarm sleeve seam. The button loop accommodates a 24-ligne button (see figures 5 and 6).

3.3.3 Figures. Figures 1 through 6 are furnished for informational purposes only. To the extent of any inconsistencies between the written document and the figures, the written document shall govern.

3.4 Materials.

3.4.1 Basic material. The basic material for the outer shell and body lining of the coat shall be water-repellent-treated polyester/cotton poplin cloth meeting the requirements of A-A-50531. The cloth shall be a plain weave and shall match the standard sample for shade, appearance, and colorfastness in the shade specified in the contract or order (see 6.2). Requirements in A-A-50531 for laundering shall not apply for the all-weather coat.

3.4.2 Removable liner materials. The insulated removable liner's material shall be constructed of a layer of batting sandwiched between a top layer and a bottom layer of taffeta material. The three layers shall be channel quilted together. The channels shall be a minimum of 5±1/2 inches wide and the stitching shall be 301 stitch type with 11 stitches per inch.

3.4.2.1 Batting. The batting material for the body of the removable liner shall be a thin, lightweight, double scrim batting composed of 65 percent polyolefin/35 percent polyester, weighing a minimum of 4.6 ounces per square yard (weight excludes scrim) and a minimum of 0.4 inch thick.

3.4.2.2 Taffeta. The taffeta shall consist of continuous filament 70 denier yarn, with a minimum of 100 yarns per inch in the warp direction and 68 yarns per inch in the filling direction, weighing 1.5±0.15 ounces per square yard, and the shade shall be a good match to the basic fabric and shall have colorfastness requirements equal to that of the basic material.

3.4.2.3 Binding material. The binding shall be cut from the taffeta material specified in 3.4.2.2 and the shade shall be a good match to the basic material and have colorfastness requirements equal to that of the basic material. Commercial binding is acceptable if the shade and physical properties are met.

3.4.2.4 Liner sleeves. When tested as specified in 4.4.1, the material for the removable liner sleeves shall meet the requirements of table II. The material shall be finished so as to provide a lustrous face and a napped back.

TABLE II. Liner sleeve material properties.

| Characteristic | Requirement |
|---|---|
| Material description | Multi-filament acetate faced, combed cotton backed, 5 or 8 harness satin weave material |
| Shade (face side) | Good match to the basic material |
| Ends per inch | 194 |
| Picks per inch | 46 |
| Weight, ounces per square yard | 6.2 |
| Breaking strength, pounds | |
| Warp | 76 |
| Filling | 75 |
| Colorfastness | |
| Gas fading | 3-4 |
| Perspiration | 3-4 |
| Dry cleaning | 3-4 |
| Crocking | |
| Dry | 4 |
| Wet | 4 |
| Dimensional stability, percent shrinkage/growth | |
| Warp | 4 |
| Filling | 2 |
| Labile sulfur | "Free" |

3.4.2.5 Hanger loop material. The hanger loop shall be made from the basic material and shall be 1/4 inch wide with a minimum breaking strength of 50 pounds.

3.4.2.6 Fusible interfacing. The interfacing shall be a nonwoven fusible material consisting of 85 percent polyester and 15 percent polyamide or 100 percent polyester, with a weight of 1.75 to 2.0 ounces per square yard, with a polyamide dot adhesive. The shade shall be charcoal or black and shall show a colorfastness to drycleaning (3 cycles) minimum rating of 3.0. For optimum fusing results, specific fuse times and settings should be obtained from the manufacturer of the non-woven fusible.

3.5 Components.

3.5.1 Slide fastener. The slide fastener for attaching the liner to the coat shall be in accordance with A-A-55634, type III, style 1 or 2, light-size with an aluminum or brass chain and a short tab pull. The shade of the tape shall be a good match to the basic material and shall be 7/16 to 9/16 inch wide. The ends of the slide fastener shall be equidistant from the bottom edge of the coat.

3.5.2 Button requirements. The buttons shall be in accordance with type I, class 1 of A-A-59994 in the styles and sizes indicated in table III. The buttons shall have a glossy finish. Unless otherwise specified (see 6.2), the button shade shall be a good match to the basic material.

TABLE III. Buttons.

| Button placement | Style | Size diameter ligne | Thickness [1/] ligne (minimum) | Total quantity per coat |
|---|---|---|---|---|
| Front and inside anchor | 20 or 21 | 45 | 7 | 8 |
| Sleeve and shoulder strap | 20 or 21 | 30 | 5 | 4 |
| Sleeve liner | 20 or 21 | 24 | 5 | 2 |
| Vent button | 20 or 21 | 24 | 5 | 1 |
| Stay buttons | 15 | 18 | 4 | 4 |
| Extra buttons | 20 or 21 | 45 | 7 | 1 |
| | 20 or 21 | 30 | 5 | 1 |

FOOTNOTE:

1/ Thickness shall be reported to the nearest full ligne.

3.5.3 Threads and gimp. The thread for seaming and stitching shall be in accordance with A-A-50199 type I, 36-45 Tex, two-ply thread for needle, bobbin, or looper or 31-35 Tex, two-ply thread for bobbin or looper. The thread for button sewing shall be in accordance with A-A-50199 type I, 61-70 Tex, two- or three-ply. The threads' finish shall be non-staining and non-flame propagating. The threads shall be treated with a permanent water repellent and the shade shall be a good match to the basic material. The threads shall have "good" colorfastness to wet drycleaning when tested in accordance with AATCC TM15. The gimp for reinforcing buttonholes shall be cotton in accordance with size no. 8, type I or II of A-A-50198. Unless otherwise specified (see 6.2), the shade shall be a good match to the basic material.

3.5.4 Belt. The belt for the women's all-weather coat shall be in accordance with A-A-55212 in the class and size appropriate to the class and size of the coat being produced. The buckle for the belt shall be in accordance with A-A-55212, and a finish or protective coating of lacquer or other finishing materials shall not be applied.

3.5.5 Seam tape. The seam tape for the topstitched seams shall be a lightweight, thermally bonded open net structure formed from pure polyamide or polyester extruded adhesive and shall weigh a minimum of 0.5 ounces per square yard and a maximum of 1.0 ounces per square yard. The polyamide adhesive shall be nylon 6, 6, 6, 11, 12 or any co-polymer blends thereof and shall be applied to the substrate by powder dot, paste dot, hot melt dot, bicomponent dot, sintered, or spunfused process. The polyester adhesive shall be polyester resin applied to the substrate by powder dot, paste dot, hot melt dot, bicomponent dot, sintered, or spunfused process. Unless otherwise specified (see 6.2), the shade of the seam tape shall be natural. For optimum fusing results, specific fuse times and settings should be obtained from the manufacturer of the fusible seam tape.

3.5.6 Labels. The coat shall have a combination identification/instruction/size label. The combination label shall be permanently attached to the bottom left facing and shall not be visible when the coat is worn. The removable liner shall have a size label caught in the neck or side seam.

3.5.6.1 Combination identification/instruction/size label (coat). The combination identification label shall be white, commercial type, sewn in label in accordance with type IV, class 14 of MIL-DTL-32075. The label's inscription shall be in bold face type and shall have a minimum font size of 10 points. The inscription's legibility, the label, and label attachment shall last the expected life of the coat. The label shall include the following information:

Coat, All-Weather, Woman's
65% Polyester/35% Cotton (Example)
Contract: SPO100-00-0-000 (Example)
XYZ Manufacturing (Example)
Dry Clean Only
After each dry cleaning, professionally retreat coat for water repellency
Size: 14R (Example)
NSN: 8410-00-000-0000 (Example)

3.5.6.2 Size label (liner). Each removable liner shall have a white, commercial type size label in accordance with type IV, class 2 of MIL-DTL-32075. The label's inscription shall have a minimum font size of 10 points. The inscription's legibility, the label, and label attachment shall last the expected life of the coat. The length (Short, Regular, Long, and Extra-Long) may be abbreviated as follows: S, R, L, or XL. The label shall be in bold face type and include the following information:

14R (Example)

3.5.6.3 Barcode label/tag. Each item shall be individually barcoded with a paper tag for personal clothing items in accordance with type VIII, class 17 of MIL-DTL-32075. The barcoding element shall be a 13-digit National Stock Number (NSN). There shall be a 12-digit Universal Product Code (UPC) assigned for all NSNs by the Government. The initials "UPC" must appear beneath the code. The barcodes for NSN and UPC shall be a medium to high density and shall be located so that they are completely visible on the item when it is folded or packaged as specified. The label's location shall cause no damage to the item.

3.6 Patterns. The Government will furnish a complete set of graded patterns, as specified (see 6.2), or sample size with grade rules. Modifications to patterns may be made, but the design and finished measurements of the finished coat and liner shall not be altered in any way. The patterns provide for a seam allowance of 1/4 inch for shoulder strap, collar, sleeve straps, and front facing. The patterns provide a seam allowance of 3/8 inch for safety stitched seams and 1/2 inch for double-lapped, double-stitched seams.

3.6.1 List of pattern parts. The pattern list in table IV is provided to ensure that the pattern set is complete.

TABLE IV. List of pattern parts.

| **Material** | **Nomenclature of pattern parts** | **Cut parts** |
|---|---|---|
| Basic material | Back | 1 |
| | Front | 1 |
| | Top sleeve | 2 |
| | Under sleeve | 2 |
| | Front facing | 1 |
| | Collar | 1 |
| | Collar stand | 1 |
| | Back vent facing | 1 |
| | Pocket facing, cord | 2 |
| | Sleeve strap | 2 |
| | Shoulder strap | 2 |
| | Top sleeve cap | 2 |
| | Pocket welt | 2 |
| | Pocket facing | 2 |
| | Front flap, right, gun patch | 1 |
| | Front flap, left, gun patch (type I only) | 1 |
| | Outside back, yoke | 1 |
| | Hanging pocket, back | 2 |
| | Hanging pocket, front | 2 |
| | Front lining, yoke | 1 |
| | Back lining, yoke | 1 |

TABLE IV. List of pattern parts – Continued.

| Material | Nomenclature of pattern parts | Cut parts |
|---|---|---|
| Lining | Top sleeve | 2 |
| | Under sleeve | 2 |
| Fusible interfacing | Front facing | 1 |
| | Shoulder strap | 2 |
| | Pocket welt | 2 |
| | Collar | 1 |
| | Collar stand | 1 |
| | Sleeve strap | 2 |
| | Top sleeve reinforcement | 2 |
| | Under sleeve reinforcement | 2 |
| | Front flap, right, gun patch | 1 |
| | Front flap, left, gun patch (type I only) | 1 |
| Taffeta (see 3.4.2.2) | Front | 1 |
| | Back | 2 |
| Liner sleeve (see 3.4.2.4) | Top sleeve | 2 |
| | Under sleeve | 2 |

3.7 Finished measurements. The finished coats and belts shall conform to the measurements in table V. The belt loops shall have dimensions as specified in 3.3.1.2.

TABLE V. Coat finished measurements (inches). [1/]

| Size | Breast, type I | Breast, type II | Back length, type I | | | Back length, type II | | | | Sleeve length, types I and II | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Short | Reg. | Long | Short | Reg. | Long | X-Long | Short | Reg. | Long | X-Long |
| X0 | 39 | -- | 41-1/2 | 43-1/4 | 45-1/4 | -- | -- | -- | -- | 16 | 17 | 18 | -- |
| 2 | 40 | -- | 41-1/2 | 43-1/2 | 45-1/2 | -- | -- | -- | -- | 16 | 17 | 18 | -- |
| 4 | 41 | 38 | 41-3/4 | 43-3/4 | 45-3/4 | 41 | 43 | 45 | -- | 16 | 17 | 18 | 19 |
| 6 | 42-1/2 | 39 | 42 | 44 | 46 | 41-1/4 | 43-1/4 | 45-1/4 | -- | 16 | 17 | 18 | 19 |
| 8 | 44 | 40 | 42-1/4 | 44-1/4 | 46-1/4 | 41-1/2 | 43-1/2 | 45-1/2 | -- | 16 | 17 | 18 | 19 |
| 10 | 45-1/2 | 41 | 42-1/2 | 44-1/2 | 46-1/2 | 41-3/4 | 43-3/4 | 45-3/4 | -- | 16 | 17 | 18 | 19 |
| 12 | 47-1/2 | 42-1/2 | 42-3/4 | 44-3/4 | 46-3/4 | 42 | 44 | 46 | -- | 16 | 17 | 18 | 19 |
| 14 | 49-1/2 | 44 | 43 | 45 | 47 | 42-1/4 | 44-1/4 | 46-1/4 | -- | 16 | 17 | 18 | 19 |
| 16 | 51-1/2 | 45-1/2 | 43-1/4 | 45-1/4 | 47-1/4 | 42-1/2 | 44-1/2 | 46-1/2 | 48-1/2 | 16 | 17 | 18 | 19 |
| 18 | 53-1/2 | 47-1/2 | 43-1/2 | 45-1/2 | 47-1/2 | 42-3/4 | 44-3/4 | 46-3/4 | 48-3/4 | 16 | 17 | 18 | 19 |
| 20 | -- | 49-1/2 | -- | -- | -- | 43 | 45 | 47 | 49 | 16 | 17 | 18 | 19 |
| 22 | -- | 51-1/2 | -- | -- | -- | -- | 45-1/4 | 47-1/4 | 49-1/4 | 16 | 17 | 18 | 19 |
| **Tol.** | ±3/4 | ±3/4 | ±1/2 | | | ±1/2 | | | | ±1/2 | | | |

FOOTNOTE:

[1/] The coat shall be placed flat on a table and measured as specified in table VI.

TABLE VI. Methods of measurement - coat.

| Measurement | Measurement description |
|---|---|
| Breast | Twice the measurement taken with the coat buttoned from folded edge to folded in line with pit of the armhole. |
| Back length, type I | The measurement taken from the center back seam from the lower edge of the undercollar to the bottom edge of the coat. |
| Sleeve length, types I and II | The measurement taken along the underarm seams from the base of the armhole to the bottom end of the sleeve. |

3.7.1 Liner measurements. The liner shall conform to the measurements in table VII.

TABLE VII. Liner finished measurements, types I and II (inches). [1/]

| Size | Breast, type I | Breast, type II | Back length, type I | | | Back length, type II | | | | Sleeve length, types I and II | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Short | Reg. | Long | Short | Reg. | Long | X-Long | Short | Reg. | Long | X-Long |
| X0 | 31-1/2 | -- | 31-1/2 | 33-1/2 | 35-1/2 | -- | -- | -- | -- | 15 | 16 | 17 | 18 |
| 2 | 33 | -- | 31-3/4 | 33-3/4 | 35-3/4 | -- | -- | -- | -- | 15 | 16 | 17 | 18 |
| 4 | 34-1/2 | 32 | 32 | 34 | 36 | 31-1/4 | 33-1/4 | 35 | -- | 15 | 16 | 17 | 18 |
| 6 | 36 | 32-1/2 | 32-1/4 | 34-1/4 | 36-1/4 | 31-1/2 | 33-1/2 | 35-1/4 | -- | 15 | 16 | 17 | 18 |
| 8 | 37-1/2 | 33 | 32-1/2 | 34-1/2 | 36-1/2 | 31-3/4 | 33-3/4 | 35-1/2 | -- | 15 | 16 | 17 | 18 |
| 10 | 39 | 34-1/2 | 32-3/4 | 34-3/4 | 36-3/4 | 32 | 34 | 35-3/4 | -- | 15 | 16 | 17 | 18 |
| 12 | 41 | 36 | 33 | 35 | 37 | 32-1/4 | 34-1/4 | 36 | -- | 15 | 16 | 17 | 18 |
| 14 | 43 | 37-1/2 | 33-1/4 | 35-1/4 | 37-1/4 | 32-1/2 | 34-1/2 | 36-1/4 | -- | 15 | 16 | 17 | 18 |
| 16 | 45 | 39 | 33-1/2 | 35-1/2 | 37-1/2 | 32-3/4 | 34-3/4 | 36-1/2 | 38-1/2 | 15 | 16 | 17 | 18 |
| 18 | 47 | 41 | 33-3/4 | 35-3/4 | 37-3/4 | 33 | 35 | 36-3/4 | 38-3/4 | 15 | 16 | 17 | 18 |
| 20 | -- | 43 | -- | -- | -- | 33-1/4 | 35-1/4 | 37 | 39 | 15 | 16 | 17 | 18 |
| 22 | -- | 45 | -- | -- | -- | -- | 35-1/2 | 37-1/4 | 39-1/4 | 15 | 16 | 17 | 18 |
| **Tol.** | ±3/4 | ±3/4 | ±1/2 | | | ±1/2 | | | | ±1/2 | | | |

FOOTNOTE:

[1/] The liner shall be placed flat on a table and measured as listed in table VIII.

TABLE VIII. Methods of measurement – removable liner.

| Measurement | Measurement description |
|---|---|
| Breast | Twice the measurement taken with the liner folded in half (down center back) and measured from front bound edge to back folded edge in line with the pit of the armhole and multiplied by two. |
| Back length, type I | The measurement taken from the center back neck edge to the bottom of the liner. |
| Sleeve length, types I and II | The measurement taken along the underarm seams from the pit of the armhole to the bottom of the sleeve. |

3.8 Construction. Unless otherwise specified (see 6.2), seam allowances shall be as specified in 3.8.1 and 3.8.1.1.

3.8.1 Seams and stitches. All joining and topstitch seams shall have 10 to 12 stitches per inch. All safety stitch, serging, or basting seams shall contain 8 to 10 stitches per inch. The width or bight of the overedge stitching shall be not less than 3/16 inch. The top and bottom edges of belt loops shall be bartacked or line tacked to coat. The back lining shall be tacked to coat at the side seams. All bartacks shall have a minimum of 28 stitches per tack and shall be a minimum of 3/8 inch in length. Raise stitching and edge stitching shall be 1/16 inch from the edge. Safety stitches shall be 3/8-inch gage. All seams shall be sewn so that no raw edges, run offs, twists, pleats, puckers, or open seams shall result. All seams shall start and finish evenly. Thread tension shall be maintained so that the stitching will not be tight or loose and the stitching shall be balanced. The ends of all seams not caught in other seams or stitching shall be backtacked not less than 1/2 inch. Liner stitching shall be as specified in 3.4.2.

3.8.1.1 Topstitching. The following components shall be finished with two rows of topstitching 1/16 inch from the edge with 1/4-inch gage utilizing lock stitching: shoulder straps, sleeve straps, collar, pocket welt, belt, top and front edges of the lapel, front edges of the coat and bottom of front facing, outside back yoke hem, and gun patch(es). The side seams and top sleeve seams shall be safety stitched and two-needle topstitched or two-needle felled, 1/16 inch from the edge with 1/4-inch gage. The front and back sleeve seam shall have a single row of 1/4-inch topstitching extending to step as marked on pattern. The left side of the back vent and the back seam shall have a single row of lockstitch 1/4 inch from the edge. The front and back lining shall have a 3/8- to 5/8-inch clean finished hem.

3.8.2 Buttonholes. The front and lapel buttonholes shall have a 1-1/4 inch cut opening and shall be horizontal eyelet end vertical square-bar type, worked over gimp, with a minimum of 70 stitches (including bartack). The horizontal buttonholes shall be positioned in accordance with the pattern marker, three on the right front, one on the left front, and one on each lapel. Each shoulder strap and sleeve strap shall have a 7/8-inch eyelet end, square-bar type buttonhole positioned with cut opening centered, ±1/8 inch, in strap and 7/8 inch from pointed end of strap.

3.9 Shade.

3.9.1 Visual shade matching. The shade and appearance shall match the standard sample (see 6.3) when tested as specified in 4.5.

3.9.2 Labile sulfur. The coat shall not contain dyes and compounds containing sulfur capable of oxidation to sulfuric acid (see table II).

3.10 Workmanship. Each finished coat and liner shall conform to the quality of product established by this document. Defects shall not exceed the established maximum Acceptance Quality Limit (AQL), as specified (see 6.2), and shall not adversely affect the serviceability, appearance, and uniformity of the product.

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.1.1 Inspection conditions. Unless otherwise specified in this specification or applicable contract or procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.2 First article inspection. First article inspection shall be performed on the coat when a first article sample is required (see 3.1.1). This inspection shall include the examinations and tests of 4.4.1 through 4.4.3 for compliance with design, configuration, workmanship, dimensional requirements, and testing as specified in this document.

4.2.1 First article samples and acceptance criteria. The first article sample size shall be as specified (see 6.2). The sample unit shall be one coat. The lot size shall be expressed in units of coats. The presence of defects exceeding specified AQLs (see 3.10 and 6.2) or failure of any testing (see 6.2), shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2, shall consist of the examinations and tests of 4.4.1 through 4.4.3.

4.3.1 Component and material sampling and acceptance criteria. Unless otherwise specified in referenced documents or procurement documents (see 6.2), sampling shall be in accordance with table IX. The component lot shall be unacceptable if one or more sample units fail to meet any examination or test requirements specified in this document (see 6.2).

TABLE IX. Material sampling.

| **Lot size** | **Sample size (sample units)** |
|---|---|
| 800 or fewer | 2 |
| 801 through 22,000 | 3 |
| 22,001 and over | 5 |

4.3.2 End item sampling and acceptance criteria. Unless otherwise specified in the contract (see 6.2), sampling for inspection shall be performed for coats in accordance with MIL-STD-1391. The sample unit shall be one finished coat. The lot size shall be expressed in units of coats. The presence of excessive defects or failure of any testing, as specified (see 6.2), shall be cause for rejection of the lot.

4.4 Inspection methods.

4.4.1 Component and material examinations and tests. Components and materials shall be assessed in accordance with the governing component requirements herein or material specifications (see 3.4.1, 3.4.2.1 through 3.4.2.3, 3.4.2.5, 3.4.2.6, and 3.5), with the exception of the liner sleeve, which shall also be assessed in accordance with table X.

TABLE X. Liner sleeve material properties verification.

| Characteristic | Requirement |
|---|---|
| Material description | AATCC TM20 |
| Weave | Visual |
| Shade (face side) | 4.5 |
| Ends per inch | ASTM D3775 |
| Picks per inch | ASTM D3775 |
| Weight, ounces per square yard | ASTM D3776/D3776M |
| Breaking strength, pounds | |
| Warp | ASTM D5034 |
| Filling | ASTM D5034 |
| Colorfastness | |
| Gas fading | AATCC TM164 |
| Perspiration | AATCC TM15 |
| Dry cleaning | AATCC TM132 |
| Crocking | |
| Dry | AATCC TM8 |
| Wet | AATCC TM8 |
| Dimensional stability, percent shrinkage/growth | |
| Warp | AATCC TM96, VI, a |
| Filling | AATCC TM96, VI, a |
| Labile sulfur | 4.4.1.1 |

4.4.1.1 Presence of labile sulfur test. Two 1.50 (±0.01) gram samples from each sample unit shall be tested. Each of the two samples shall be cut into very small pieces and placed into separate test tubes. An acidic tin(II) chloride (stannous chloride) solution shall be prepared by dissolving 100 grams $SnCl_2$ (American Chemical Society [ACS] grade) in 100 milliliters of 35 percent hydrochloric acid (ACS grade) and adding the mixture to 50 milliliters of distilled water. A lead acetate solution shall be prepared by dissolving 5.0 grams of lead acetate (chemically pure reagent grade) in a small amount of distilled water and diluting to a total of 100 milliliters. If the solution is not clear, a few drops (one at a time) of glacial acetic acid shall be added until the solution is clear. The samples shall be submersed in the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride, and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

a. Free – The filter paper shows no discoloration or staining of any kind.

b. Slight – The filter paper shows a light tan to light brown discoloration stain.

c. Moderate – The filter paper shows a dark brown discoloration stain.

d. Severe – The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail.”

4.4.2 In-process examination. Visual and dimensional examinations shall be made at any point during the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with the requirements specified in section 3. Materials and components that contain defects and damages, as defined in the visual examination of 4.4.3, shall be removed from production. Any in-process nonconformance remaining in the finished trousers shall be classified as a defect in accordance with the visual examination of 4.4.3.

4.4.3 Visual examination. Sampling and visual examination of the end item for defects in shade, design, materials, construction, workmanship, and marking shall be performed in accordance with MIL-STD-1391. Defects in MIL-STD-1391 associated with layering of the front opening of the coat should substitute "right over left" for "left over right". Model form requirements in MIL-STD-1391 shall be ignored. Defects shall not exceed the AQLs specified in the contract or order (see 6.2).

4.5 Visual shade matching. Samples shall be viewed using AATCC EP9, option C (see 6.4), with sources simulating artificial daylight D65 illuminant (see 6.4.1) with a color temperature of 6,500±200 Kelvin (K) and illumination of 100±20 foot candles and under incandescent A illuminant with a color temperature of 2,856±200K.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coat is intended to be worn by female personnel of the United States Marine Corps, as specified by uniform regulations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. NSNs.

c. Types, classes, and sizes required (see 1.2).

d. When first article inspection is required (see 3.1).

e. When conformance inspection is required (see 3.1.2 and 4.3).

f. Shade standard information (see 3.4.1 and 6.4.2).

g. Button shade, if other than specified (see 3.5.2).

h. Thread shade, if other than specified (see 3.5.3).

i. The shade of components, if other than specified (see 3.5.5).

j. Applicable Government pattern references, with pattern date (see 3.6).

k. Seam allowances, if other than as specified (see 3.8).

l. AQLs (see 3.10).

m. First article sampling and acceptance criteria (see 4.2.1).

n. Component and material sampling and acceptance criteria (see 4.3.1 and 4.4.3).

o. End item sampling and acceptance criteria (see 4.3.2).

p. Component and material examination and tests (see 4.4.1).

q. Packaging requirements (see 5.1).

6.3 Information requests. Requests for information such as purchase descriptions, patterns, drawings, and standard shade samples of cloth can be made by completing and submitting the Defense Logistics Agency Troop Support's Clothing and Textile Specification/Drawing/Pattern Request Form online at https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Requests to use equivalent materials or components or to make changes to the pattern should be sent to the contracting officer for approval by the military services.

6.4 Visual shade matching. In 2017, option A of AATCC EP 9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be, “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.”

6.4.1 Use of D75 illumination source. The use of D75 illuminant with a color temperature of 7,500±200K and illumination of 100±20 foot candles in lieu of the specified D65 illuminant (see 4.5) is permitted.

6.4.2 Shade criticality. Some items may be deemed “non-shade-critical” by the contracting agency and alternative shade standards or information regarding shade may be specified in the contract or order (see 6.2). It is recommended that manufacturers refer to the contract or order to determine the criticality of the shade matching or alternate shade standards that are acceptable.

6.5 Subject term (key word) listing.

Liner
Poplin
Water-resistant

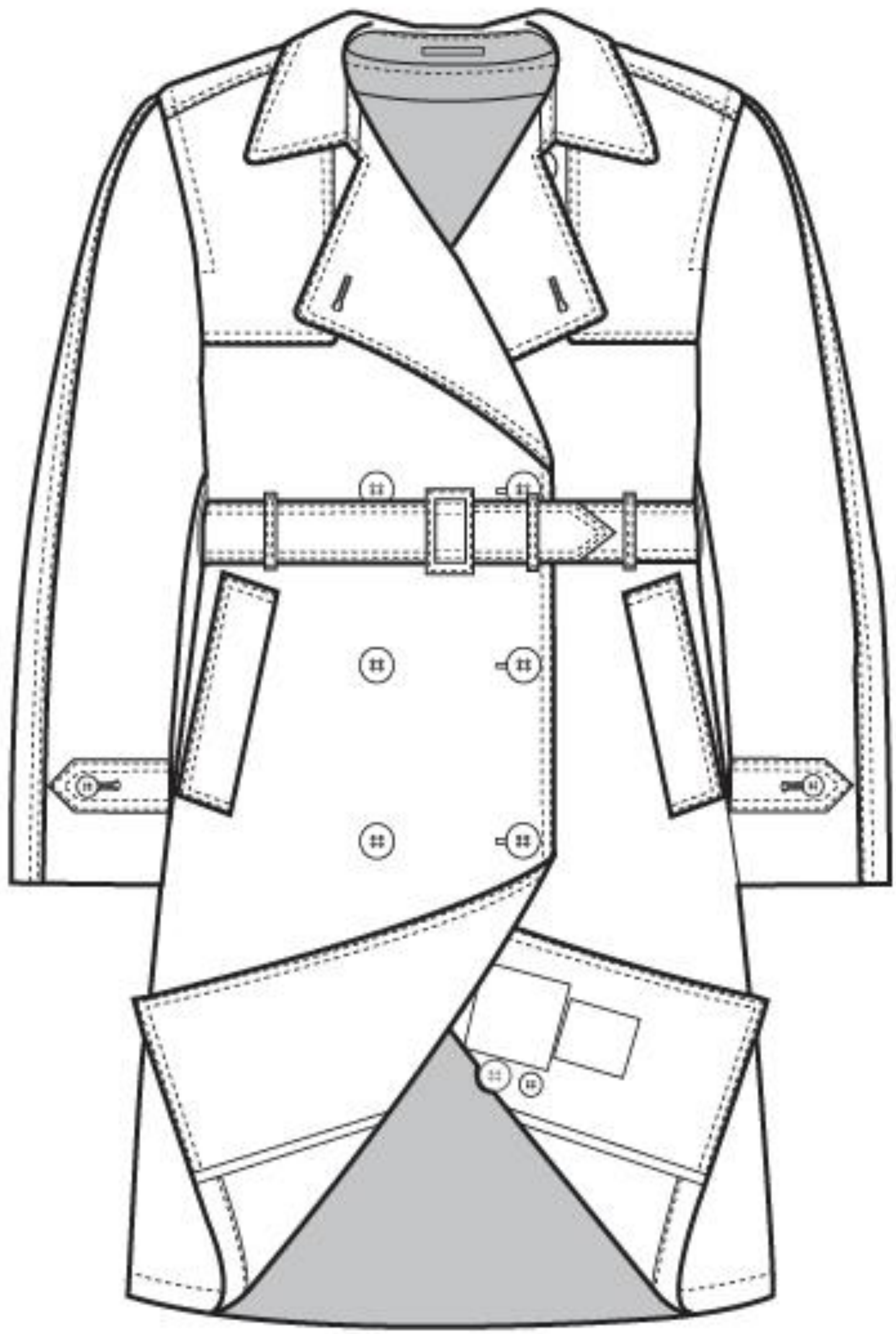

FIGURE 1. <u>All-weather coat front, type I (Marine Corps)</u>.

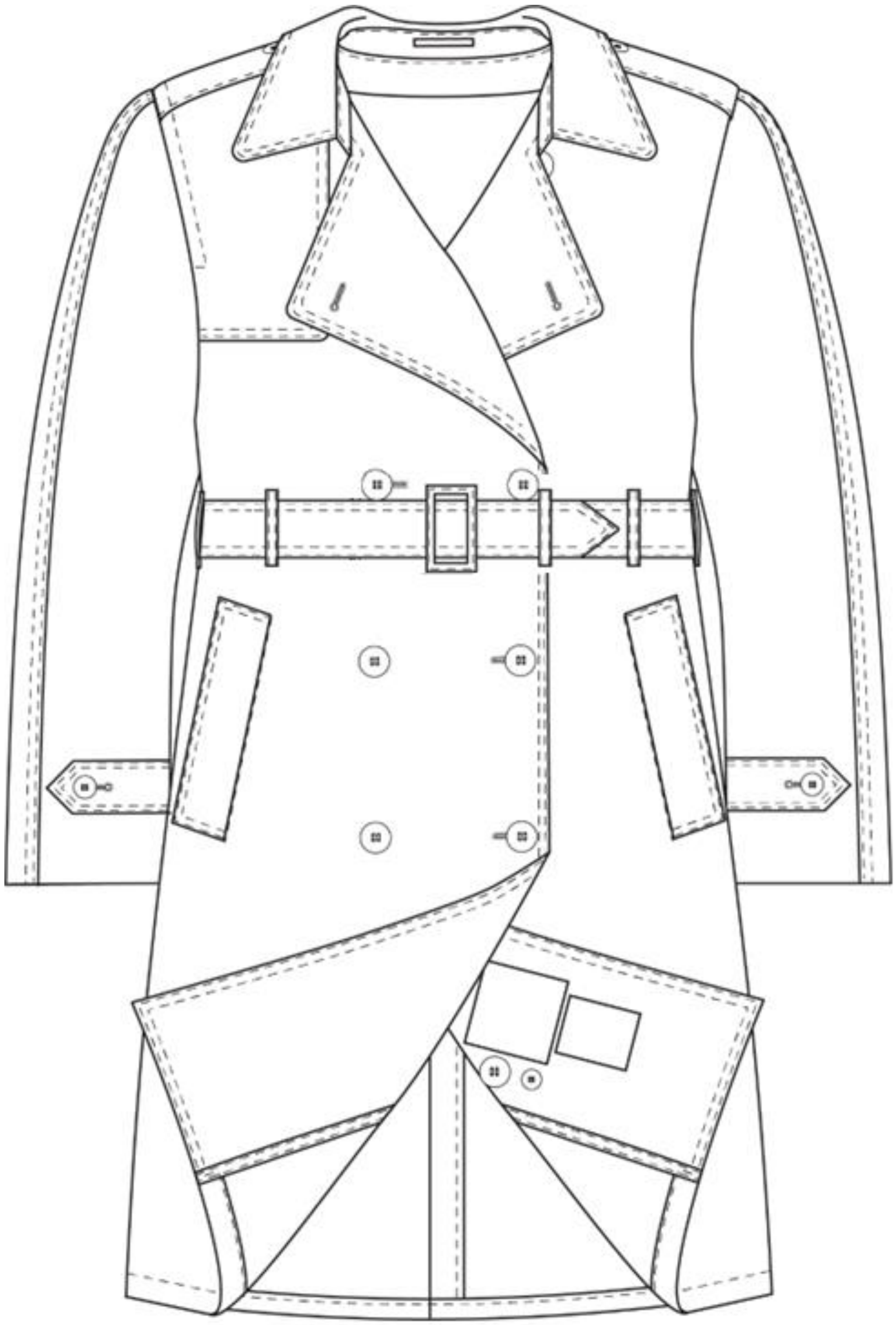

FIGURE 2. All-weather coat front, type II (Army and Navy).

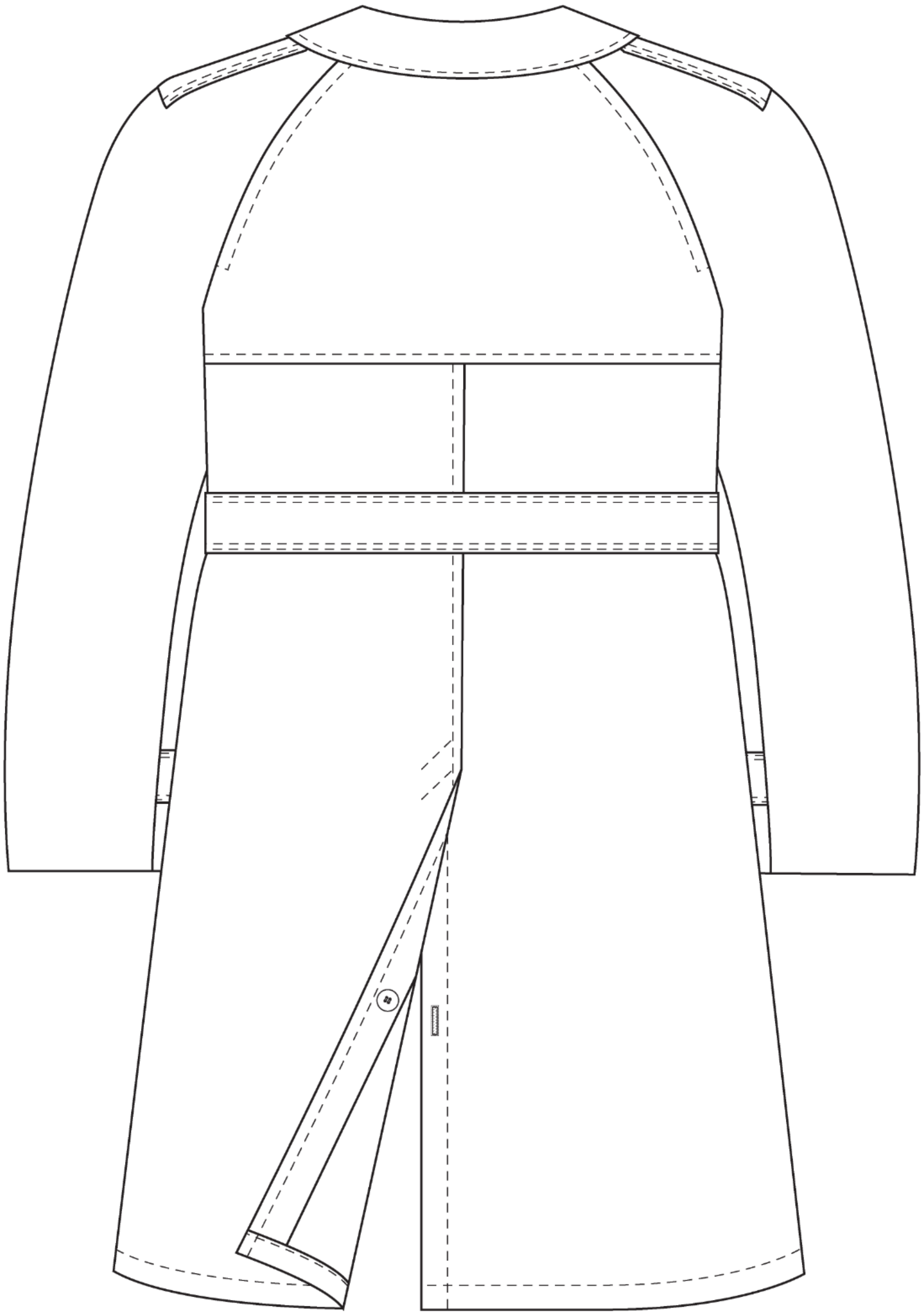

FIGURE 3. All-weather coat back, all types.

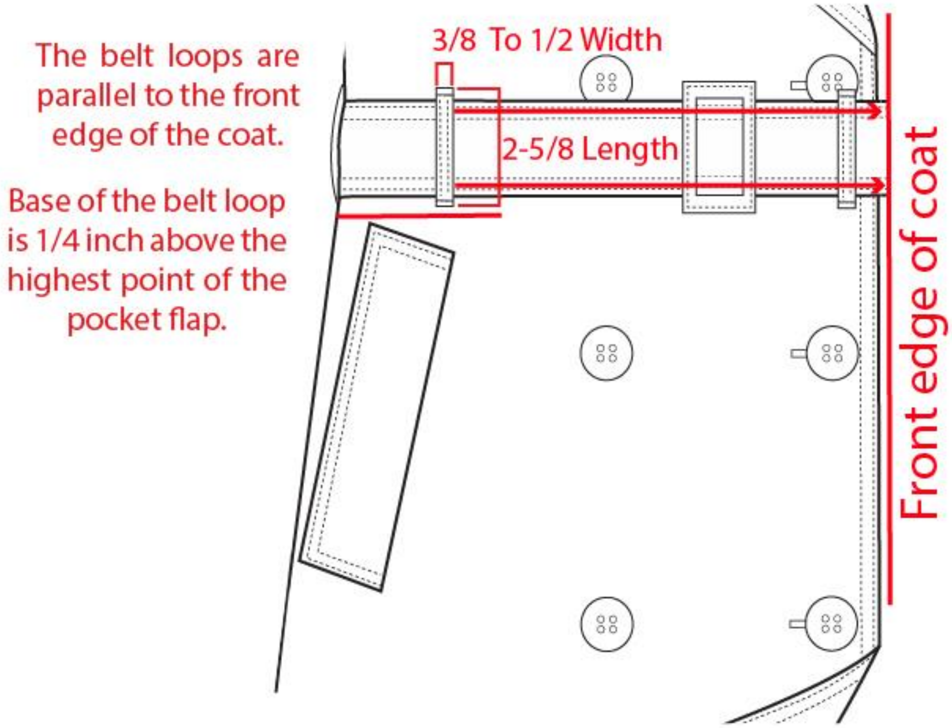


FIGURE 4. All-weather coat front belt loop placement, all types.

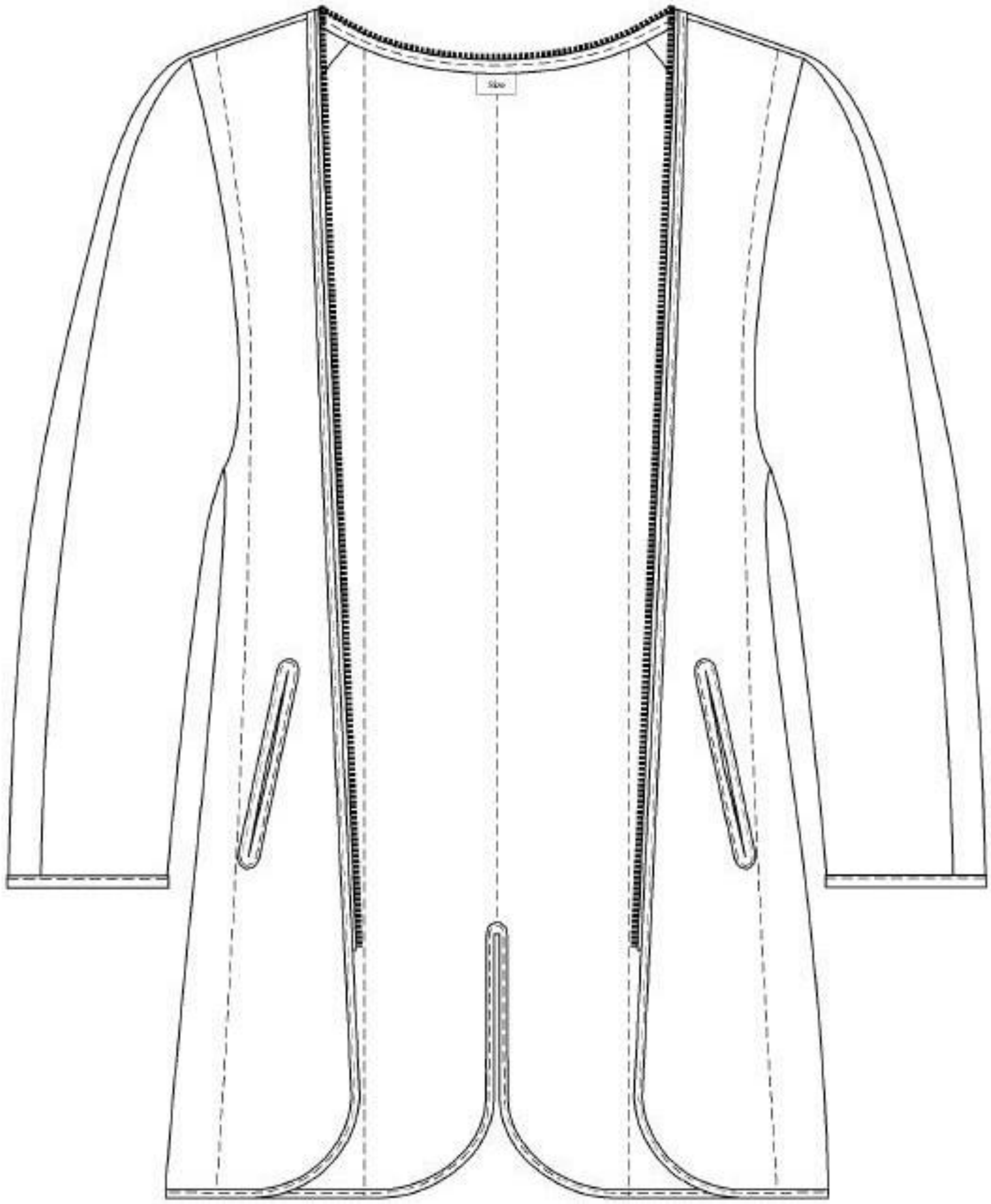

FIGURE 5. Removable liner front, all types.

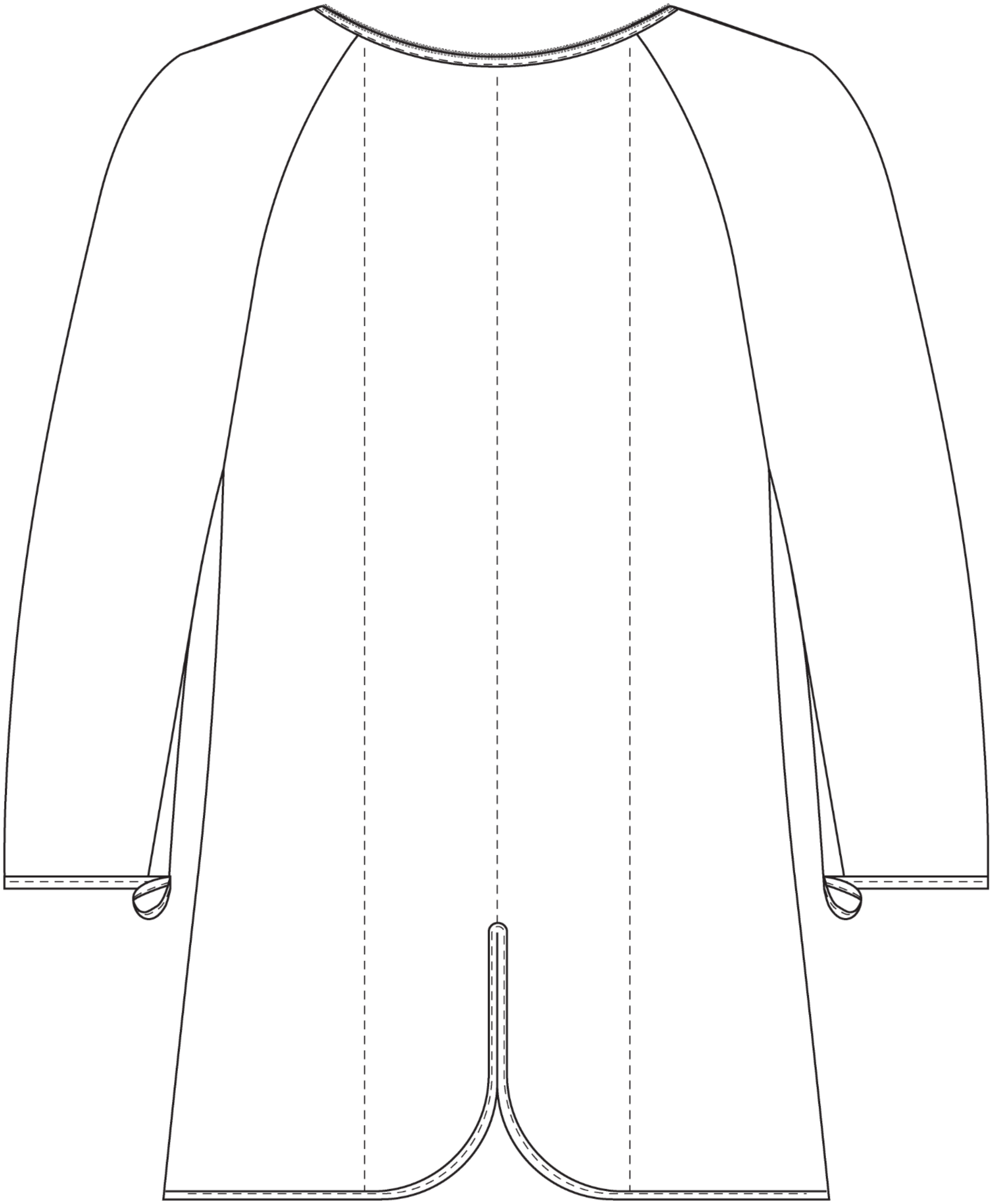

FIGURE 6. Removable liner back, all types.

CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC

Preparing activity:
Navy – MC
(Project 8410-2024-001)

Review activities:
Army – AV, MI
Navy – CG1, NU
Air Force – 11
DLA – CT

Civil agency:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.